Lab 6 实验报告

实验要求

支持中断与异常。

实验过程

这一小节中，所有 AI 帮助完成的部分都进行了标注

参考

- AI(DeepSeek)

Step 0

调整了原本的异常检测逻辑，从 `ex_stage` 中移除，形成了单独的模块，完成了相关逻辑。

Step 1

将 `trap` 模块接入了流水线，并调整了部分流水线逻辑。

调试过程

这一小节中，所有 AI 帮助完成的部分都进行了标注

问题 1

`load` 和 `store` 相关异常检测失败。

解决 1

发现是检测信号写错了。

问题 2

中断检测失败。

解决 2

发现是对 `mip` 的逻辑理解错误，并且在 `mret` 后立刻检测中断时，会导致在当前指令执行结束前就开始检测，出现错误。

因此强制要求在 `mret` 指令执行结束前不能执行中断检测。避免特权级别的改变产生影响。

问题 3

`lab3` 中的数据转发问题暴露。

解决 3

具体来说，`ex` 阶段的指令会比 `id` 阶段的指令提前一个周期进入流水线寄存器，导致转发出现问题。因此要求 `ex_mem` 寄存器和 `id_ex` 寄存器的地址必须不同时才转发。

问题 4

`lab5` 会触发非法指令。

解决 5

我认为是 `lab5` 的指令集中本身就包含一个非法指令。因为我取指时的 `pc` 并没有问题，而且也不是 `M-mode`，因此不会进行 `mmu` 的地址转换。

因此，取指方面应该没有问题，应该就是 `lab5` 中有一个非法指令。

以下是测试结果：

lab6

```
Single test passed.
Run sys-test
trap here, epc 8000600c, cause 8
Test ecall_u [OK]
trap here, epc 8000608c, cause 8
trap here, epc 80006028, cause 0
Test instr_misalign [OK]
trap here, epc 8000608c, cause 8
trap here, epc 80006040, cause 4
Test load_misalign [OK]
trap here, epc 8000608c, cause 8
trap here, epc 80006050, cause 6
Test store_misalign [OK]
trap here, epc 8000608c, cause 8
trap here, epc 800060b4, cause 8000000000000007
Test timer_intr [OK]
trap here, epc 8000608c, cause 8
trap here, epc 80006090, cause 8000000000000003
Test software_intr [OK]
trap here, epc 8000607c, cause 8
Timer interrupt in test_trap, this should happen 50 times.
Timer interrupt in test_trap, this should happen 50 times.
Timer interrupt in test_trap, this should happen 50 times.
Timer interrupt in test_trap, this should happen 50 times.
```

lab5

```
kinit ok
procinit ok
trapinit ok
plicinit ok
userinit ok
sh: 1: spike-dasm: not found
```

以下是出错的地址、指令以及调试信息：

```
commit inst [9]: pc 0080001d60 inst 12000073 wen 0 dst 00000000 data 0000000000000000 <--
```

```
mstatus different at pc = 0x0080001d60, right= 0x0000000000000080, wrong = 0x0000000000001800
 mcause different at pc = 0x0080001d60, right= 0x0000000000000008, wrong = 0x0000000000000002
   mepc different at pc = 0x0080001d60, right= 0x00000007ffff0008, wrong = 0x0000000080001d60
```

lab4

```
[src/cpu/cpu-exec.c:390,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x000000008001fff8
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 2,080 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 32,766
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 15,752,884 instr/s
```

lab3

```
[TEST] AES-128 test vector Encrypt
[PASS] AES-128 test vector Encrypt passed
[TEST] AES-128 test vector Decrypt
[PASS] AES-128 test vector Decrypt passed
[PASS] AES correctness checks passed
[RNG] heavy_rng_mod result=0, time(us)=347300
[RNG] heavy_rng_mod it/s=2881
[src/cpu/cpu-exec.c:390,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x0000000080000030
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 60,026 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 1,243,571
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 20,717,205 instr/s
```

lab2

```
[src/cpu/cpu-exec.c:390,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x000000008001fffc
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 2,140 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 32,767
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 15,311,682 instr/s
```

lab1

```
[src/cpu/cpu-exec.c:390,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x0000000080010004
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 939 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 16,385
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 17,449,414 instr/s
```